GitHub – Basic Concepts

Simple Presentation Notes for Beginners

Prepared for: Beginner-level GitHub Learning

1. What is GitHub?

GitHub is an online platform used to store, manage, share, and collaborate on software projects and other files. It uses Git for version control.

2. What is Git?

Git is a version control system. It records changes made to files so that users can track their work and return to earlier versions when needed.

3. Repository

A repository, or repo, is a project space on GitHub. It can contain code, documents, images, notes, and other project files.

4. Commit

A commit is a saved record of changes in a Git repository. A good commit message briefly explains what was changed.

5. Branch

A branch is a separate line of development. It allows you to work on a feature or change without directly changing the main branch.

6. Push and Pull

Push sends local commits to a remote GitHub repository. Pull gets the latest changes from the remote repository to your local project.

7. Pull Request

A pull request is a way to propose changes from one branch to another. Team members can review the changes, discuss them, and merge them.

8. Clone

Cloning creates a local copy of a GitHub repository on your computer. You can then work on the project locally.

9. README.md

A README file explains a repository. It commonly contains the project title, purpose, features, installation steps, usage, and learning information.

10. GitHub Workflow

A simple workflow is:

- Create or open a repository
- Create or edit files
- Stage changes
- Commit changes
- Push changes to GitHub
- Create a branch when needed
- Use a pull request for review

11. Why Use GitHub?

- Keeps project history and versions
- Makes collaboration easier
- Provides online backup for repositories
- Helps organize projects
- Useful for building a portfolio
- Supports code review and open-source collaboration

12. Important Git Commands

Command	Purpose

git init	Initialize a Git repository
git add .	Stage changes
git commit -m "message"	Save changes with a message
git push	Upload commits to GitHub
git pull	Download and integrate latest changes
git clone <URL>	Copy a remote repository locally

Conclusion

GitHub is an important platform for learning, managing, sharing, and collaborating on projects. Understanding repositories, commits, branches, push, pull, and pull requests gives a strong foundation for using GitHub effectively.